

14. An atom of an element has electronic configuration 2, 3. It forms a compound with an element having the electronic configuration 2, 7. Identify the false statement regarding the compound.
- (a) It possesses contracted octet.
 - (b) It can act as an electron acceptor.
 - (c) There are three lone pairs on a non-metal atom.
 - (d) It is an unstable molecule.
15. Identify the molecule having ionic bond, covalent bond and dative bond.
- (a) Ammonium hydroxide
 - (b) Ammonium chloride
 - (c) Blue vitriol
 - (d) Ammonium carbonate

Space for rough work

Assessment Test IV

Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

1. **Assertion (A):** Carbon shows tetravalency in the compounds.

Reason (R): Atoms attain the nearest inert gas configuration during bonding.

- (a) Both A and R are true and R is the correct explanation of A.
 (b) Both A and R are true, but R is not the correct explanation of A.
 (c) A is true and R is false.
 (d) A is false and R is true.

2. **Assertion (A):** HF has a greater boiling point than H_2O .

Reason (R): HF has a stronger hydrogen bonding than water.

- (a) Both A and R are true and R is the correct explanation of A.
 (b) Both A and R are true, but R is not the correct explanation of A.
 (c) A is true and R is false.
 (d) A is false and R is true.

3. Arrange the following atomic numbers in the decreasing order of ease of formation of ionic compounds with a metal 'M'.

- (A) 8 (B) 7 (C) 17 (D) 9
 (a) DBAC (b) CABD (c) CDAB (d) CBDA

4. A metal 'M' forms ionic compounds with the following non-metals.

- (A) Bromine (B) Oxygen (C) Iodine
 (D) Fluorine (E) Sulphur

Arrange these non-metals in the decreasing order of the strength of ionic bonds.

- (a) DBEAC (b) CAEBD
 (c) BDEAC (d) CBEAD

5. Match the statements of Column A with those of Column B

Column A	Column B
(i) Ammonium ion	(A) Delocalized electrons
(ii) Nitrogen molecule	(B) Crystal lattice
(iii) Mercury liquid	(C) Dative bond
(iv) Magnesium oxide	(D) Triple covalent bond
(v) Ammonia liquid	(E) Hydrogen bonding

Space for rough work

- (a) (i) $\rightarrow$ (C); (ii) $\rightarrow$ (D); (iii) $\rightarrow$ (D); (iv) $\rightarrow$ (B); (v) $\rightarrow$ (D)
(b) (i) $\rightarrow$ (D); (ii) $\rightarrow$ (A); (iii) $\rightarrow$ (D); (iv) $\rightarrow$ (B); (v) $\rightarrow$ (C)
(c) (i) $\rightarrow$ (D); (ii) $\rightarrow$ (D); (iii) $\rightarrow$ (A); (iv) $\rightarrow$ (B); (v) $\rightarrow$ (C)
(d) (i) $\rightarrow$ (C); (ii) $\rightarrow$ (D); (iii) $\rightarrow$ (A); (iv) $\rightarrow$ (B); (v) $\rightarrow$ (D)

6. Which of the following occurs due to redox reaction?
- (a) Liberation of H_2 gas on reaction of metal with dilute acid.
(b) Burning of Mg ribbon in air.
(c) Formation of ferrous sulphide when iron is heated with sulphur.
(d) Precipitation reaction.
7. In which of the following compounds do the metal and the non-metal possess electrovalencies of 3 and 2, respectively?
- (a) M_2X (b) M_3X_2 (c) M_2X_3 (d) MX_3
8. A metal has 'M' shell as valence shell with one electron in it. A non-metal exhibits an electrovalence of 3 and has only two shells in its atom. Which combination of inert gases does the ions so formed attain?
- (a) Ne, Ar (b) Ar, Ar
(c) Ar, Kr (d) Ne, Ne
9. An element has the electronic configuration 2, 7. If it forms a diatomic molecule with its own element, how many lone pairs can be present on each atom after bond formation?
- (a) 0 (b) 2 (c) 3 (d) 1
10. Two elements belonging to the same group form a covalent compound. Identify the probable atomic numbers of the elements.
- (a) 16, 8 (b) 20, 1 (c) 16, 9 (d) 20, 9
11. Iron is a hard metal whereas sodium is a soft metal. Identify the reason.
- (a) Low ionization potential value of sodium
(b) High ionization potential value of sodium
(c) More number of free electrons in sodium
(d) More number of free electrons in iron
12. Hydrogen bonding imparts some characteristic properties to substances. Identify the property which does not come under that category.
- (a) Relatively higher boiling point of water
(b) Dissolution of glucose in water
(c) Existence of water in liquid state
(d) Conduction of electricity

13. How many bond pairs are present in the Lewis dot diagram of ethylene molecule?
 (a) 2 (b) 4 (c) 5 (d) 6
14. An atom of an element has electronic configuration 2, 8, 6. Identify the number of electrons around the central atom of the compound formed with an element having the electronic configuration 2, 7.
 (a) 8 (b) 10 (c) 12 (d) 14
15. Which of the following bonds is present in hydronium ion?
 (a) Ionic bond (b) Covalent bond
 (c) Dative bond (d) None of these

Space for rough work

Answer Keys

Assessment Test I

1. (c) 2. (d) 3. (b) 4. (a) 5. (b) 6. (c) 7. (a) 8. (d) 9. (c) 10. (d)
 11. (d) 12. (d) 13. (c) 14. (d) 15. (a)

Assessment Test II

1. (b) 2. (d) 3. (c) 4. (d) 5. (c) 6. (d) 7. (b) 8. (c) 9. (a) 10. (b)
 11. (a) 12. (a) 13. (b) 14. (a) 15. (b)

Assessment Test III

1. (c) 2. (a) 3. (d) 4. (d) 5. (d) 6. (c) 7. (d) 8. (d) 9. (b) 10. (a)
 11. (b) 12. (d) 13. (b) 14. (b) 15. (a)

Assessment Test IV

1. (b) 2. (a) 3. (a) 4. (c) 5. (a) 6. (c) 7. (b) 8. (d) 9. (c) 10. (c)
 11. (a) 12. (d) 13. (b) 14. (a) 15. (a)

Mole Concept, Stoichiometry and Behaviour of Gases

5



Reference: Coursebook - IIT Foundation Chemistry Class 9; Chapter - Mole Concept, Stoichiometry and Behaviour of Gases; Page number - 5.1–5.23

Assessment Test I

Time: 30 min.

Directions for questions from 1 to 15: Select the correct answer from the given options.

Space for rough work

1. Which among the following equations does not comply with Charles' law?

- (a) $\frac{V}{T} = \text{constant}$ (b) $\frac{V_1}{V_2} = \frac{T_1}{T_2}$
(c) $\frac{V_1}{T_1} = \frac{V_2}{T_2}$ (d) $V_1 T_1 = V_2 T_2$

2. **Assertion (A):** 1 mole of oxygen always contains Avogadro number of molecules.

Reason (R): 22.4 L of oxygen always contains Avogadro number of molecules.

- (a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true and R is false.
(d) A is false and R is true.

3. Arrange the following in the decreasing order of the total number of atoms present in one mole each of:

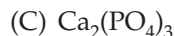
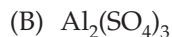
- (A) Sulphuric acid (B) Aluminium oxide
(C) Potassium dichromate (D) Ferrous sulphate
(E) Ammonium bicarbonate
(a) ECADB (b) CAEDB
(c) ECBAD (d) CEADB

4. **Assertion (A):** A cooking gas cylinder, emptied under normal conditions, if kept in hot water, can still be used for some more time.

Reason (R): The pressure exerted by a gas is proportional to the number of collisions of gas molecules with the walls of the container.

- (a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true and R is false.
(d) A is false and R is true.

5. Identify the incorrect formulae among the following.



(A) DE

(b) AC

(c) AE

(d) CD

6. A gas was transferred from one-litre cylinder into another cylinder of different volume at the same temperature. Then, the pressure exerted by the gas increased by 20%. Identify the volume of the new cylinder.

(a) 480 mL

(b) 1170 mL

(c) 833 mL

(d) 607 mL

7. Arrange the following conditions of pressure and temperature in the increasing order of volumes occupied by one mole of a gas.

(A) 600 mm and 200 K

(B) 400 mm and 100 K

(C) 600 mm and 100 K

(D) 400 mm and 200 K

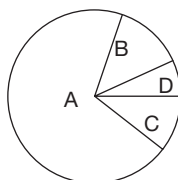
(a) CBAD

(b) BCAD

(c) CADB

(d) BACD

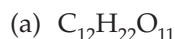
8. The following pie chart represents the percentage composition by weight of calcium bicarbonate. Identify the components 'A', 'B', 'C' and 'D' indicated in the pie chart.

(a) $\text{A} \rightarrow \text{Ca}$; $\text{B} \rightarrow \text{O}$; $\text{C} \rightarrow \text{C}$; $\text{D} \rightarrow \text{H}$ (b) $\text{A} \rightarrow \text{O}$; $\text{B} \rightarrow \text{Ca}$; $\text{C} \rightarrow \text{C}$; $\text{D} \rightarrow \text{H}$ (c) $\text{A} \rightarrow \text{Ca}$; $\text{B} \rightarrow \text{C}$; $\text{C} \rightarrow \text{H}$; $\text{D} \rightarrow \text{O}$ (d) $\text{A} \rightarrow \text{O}$; $\text{B} \rightarrow \text{C}$; $\text{C} \rightarrow \text{Ca}$; $\text{D} \rightarrow \text{H}$

9. x grams of a diatomic gas occupies 5.6 L volume at STP. Identify the incorrect statement regarding the gas.

(a) The sample of gas contains 1.5×10^{23} molecules.(b) The total number of atoms will be 3×10^{23} .(c) If the gas is oxygen, then x is equal to 8 g.(d) The vapour density of the gas would be $4x$.

10. The empirical formula mass of a compound is one-sixth of its molecular mass. What is the molecular formula corresponding to that substance?



Space for rough work

11. A trivalent metallic oxide contains 68.4% by weight of metal. What could be the atomic weight of the metal?
 (a) 27 (b) 39 (c) 52 (d) 152
12. 70 g of a mixture of sodium carbonate and sodium bicarbonate is taken and subjected to heating until there is no further loss in weight. The heating was followed by condensation. The loss in weight of the sample was 15.7% of the initial weight of the mixture. Find out the amount of the sodium carbonate in the mixture.
 (a) 42 g (b) 53 g (c) 28 g (d) 64 g
13. When 1.225 kg of potassium chlorate was subjected to decomposition, the oxygen evolved in the reaction occupied a volume of 224 L. What would be the percentage purity of potassium chlorate?
 (a) 82% (b) 30% (c) 66% (d) 25%
14. Match the statements of Column A with those of Column B.

Column A (Formula of Hydrocarbon)	Column B (Volume Ratio of CO_2 and H_2O at STP)
(i) C_2H_2	(A) 1:1
(ii) CH_4	(B) 2:1
(iii) C_2H_4	(C) 2:3
(iv) C_2H_6	(D) 1:2

- (a) (i) $\rightarrow$ (B); (ii) $\rightarrow$ (D); (iii) $\rightarrow$ (A); (iv) $\rightarrow$ (C)
 (b) (i) $\rightarrow$ (B); (ii) $\rightarrow$ (A); (iii) $\rightarrow$ (D); (iv) $\rightarrow$ (C)
 (c) (i) $\rightarrow$ (C); (ii) $\rightarrow$ (D); (iii) $\rightarrow$ (A); (iv) $\rightarrow$ (B)
 (d) (i) $\rightarrow$ (C); (ii) $\rightarrow$ (A); (iii) $\rightarrow$ (D); (iv) $\rightarrow$ (B)
15. Two moles of a hydrocarbon for complete combustion consume 13 moles of oxygen. When 1 mole of it is subjected to complete combustion, the number of moles of water vapour formed is one mole more than the number of moles of CO_2 formed. Identify the formula of hydrocarbon.
 (a) C_4H_{10} (b) C_5H_{12}
 (c) C_5H_{10} (d) C_4H_8

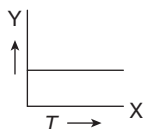
Assessment Test II

Time: 30 min.

Space for rough work

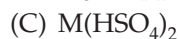
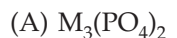
Directions for questions from 1 to 15: Select the correct answer from the given options.

1. A graph is given below by taking Kelvin temperature on the 'X' axis and another variable on the 'Y' axis. From the nature of the graph, identify the variable taken on the 'Y' axis.



- (a) V (b) VT (c) $\frac{V}{T}$ (d) $\frac{1}{V}$
2. **Assertion (A):** Two moles of a given mass of a gas occupy a volume of 44.8 L at STP.
Reason (R): Volume of a given mass of a gas depends on temperature and pressure.
- (a) Both A and R are true and R is the correct explanation of A.
 (b) Both A and R are true, but R is not the correct explanation of A.
 (c) A is true and R is false.
 (d) A is false and R is true.
3. Arrange the following in the decreasing order of number of oxygen atoms.
- (A) 0.5 moles of sulphuric acid (B) 1 mole of water
 (C) 0.25 moles of nitric acid (D) 2 moles of hydrogen peroxide
 (E) 1 mole of ferric oxide
- (a) DABEC (b) BAECD
 (c) DEABC (d) BCDEA
4. **Assertion (A):** A balloon filled with air and tightly tied, when hung close to the source of heat, bursts after some time.
Reason (R): The pressure exerted by the gas is inversely proportional to the temperature of the gas.
- (a) Both A and R are true and R is the correct explanation of A.
 (b) Both A and R are true, but R is not the correct explanation of A.
 (c) A is true and R is false.
 (d) A is false and R is true.
5. A metal 'M' forms a compound of the formula M_2HPO_4 . Identify the correct formulae of the compounds formed by the same metal radical from the following.

Space for rough work



(a) ABD

(b) CDE

(c) BDE

(d) BCE

6. What is the percentage change in pressure exerted by a gas when it is transferred from a 6-L cylinder to an 8-L cylinder at constant temperature?

(a) -75%

(b) +75%

(c) +25%

(d) -25%

7. Arrange the following closed reaction vessels containing equal number of moles of different gases, in the decreasing order of pressure exerted by them.

(A) 200 mL container at 200 K

(B) 600 mL container at 300 K

(C) 200 mL container at 400 K

(D) 600 mL container at 500 K

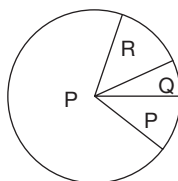
(a) DBAC

(b) CADB

(c) CDBA

(d) DCBA

8. The following pie chart represents the percentage composition of sodium bicarbonate. Identify the components 'P', 'Q', 'R' and 'S' indicated in the pie chart.

(a) $P \rightarrow O$; $Q \rightarrow C$; $R \rightarrow H$; $S \leftrightarrow Na$ (b) $P \rightarrow Na$; $Q \rightarrow H$; $R \rightarrow C$; $S \rightarrow O$ (c) $P \rightarrow O$; $Q \rightarrow H$; $R \rightarrow C$; $S \rightarrow Na$ (d) $P \rightarrow C$; $Q \rightarrow H$; $R \rightarrow Na$; $S \rightarrow O$

9. The vapour density of a gas is x . What would be the number of molecules associated with $3x/2$ g of the gas?

(a) 4.5×10^{23} (b) 3×10^{23} (c) 6×10^{23} (d) 9×10^{23}

10. Which among the following pairs has the same empirical formula?

(a) Glucose and acetic acid

(b) Ethyl alcohol and sucrose

(c) Ethyl alcohol and acetic acid

(d) Ethene and acetylene

11. A trivalent metal oxide contains 30% oxygen. Find the gram atomic mass of the metal present in the oxide.
- (a) 72 g (b) 160 g
(c) 112 g (d) 56 g
12. A mixture of equal amount of calcium carbonate and sodium carbonate was heated till there is no further loss in weight. If the gaseous product obtained occupies a volume of 5.6 L at STP, find out the initial weight of the mixture.
- (a) 100 g (b) 50 g
(c) 75 g (d) 25 g
13. What is the extra amount of potassium chlorate to be taken to get 134.4 L of oxygen at STP by the decomposition of 80% pure potassium chlorate?
- (a) 490 g (b) 612.5 g
(c) 245 g (d) 122.5 g
14. Match the statements of Column A with those of Column B.

Column A (Formula of Hydrocarbon)	Column B (No. of Moles of Oxygen Consumed per Mole of Hydrocarbon)
(i) CH_4	(A) 3.5
(ii) C_2H_6	(B) 2
(iii) C_2H_2	(C) 3
(iv) C_2H_4	(D) 2.5

- (a) (i) $\rightarrow$ (B); (ii) $\rightarrow$ (D); (iii) $\rightarrow$ (A); (iv) $\rightarrow$ (C)
 (b) (i) $\rightarrow$ (B); (ii) $\rightarrow$ (A); (iii) $\rightarrow$ (D); (iv) $\rightarrow$ (C)
 (c) (i) $\rightarrow$ (C); (ii) $\rightarrow$ (D); (iii) $\rightarrow$ (A); (iv) $\rightarrow$ (B)
 (d) (i) $\rightarrow$ (C); (ii) $\rightarrow$ (A); (iii) $\rightarrow$ (D); (iv) $\rightarrow$ (B)
15. One mole of a hydrocarbon consumed 4.5 moles of oxygen for complete combustion and gave equal volumes of CO_2 and water vapour. Identify the formula of the hydrocarbon.
- (a) C_3H_4 (b) C_2H_4
(c) C_3H_8 (d) C_3H_6